

SUMMARY

Security engineering leader with 15 years of experience spanning software and systems engineering, applied security research, hands-on security assessments, and team leadership. PhD-trained researcher who evolved from discovering security flaws to advising and leading technical teams. Delivered hundreds of security assessments to clients ranging from startups to Fortune 500; mentored engineers; grew teams; and led cross-team initiatives to increase sales, retain customers, and streamline operations. Currently serving as a Fractional CTO and seeking an in-house security leadership role where I can apply lessons from hundreds of assessments toward bootstrapping and maturing software development and security programs that enable engineering teams long-term.

EXPERIENCE

Cloud of Witnesses Project

Fractional CTO

August 2025 - Present

- Co-founded a nonprofit tech platform for sharing and visualizing personal stories within communities of faith.
- Defined initial product vision, architecture, roadmap, development plans, and implemented the MVP.
- Researched and integrated third-party systems to enable nonprofit growth (e.g., donation management, marketing campaigns).

Trail of Bits

Principal Security Engineer & Director of Consulting

September 2022 - August 2025

Senior technical leadership bridging hands-on delivery with team development and business growth

- Led secure code reviews and threat models (Tech Lead) for numerous client engagements, managing teams of 2-6 engineers per project.
- Built internal tooling, SAST rules (semgrep), and standardized reporting practices and security finding templates.
- Advanced the AppSec team (growth and maturity) through coaching and creating personal growth plans aligned to business goals.
- Designed and implemented strategic Quarterly Business Review (QBR) playbook that drove VIP account expansion.
- Created internal service definitions that enabled sales and marketing collateral, enabling proactive client outreach and faster deal closure.
- Led initiatives to streamline AppSec pre-sales, scoping, and SOW processes, cutting turnaround times and improving delivery.
- Served as Interim Head of Assurance Practice (June-Nov 2023) during leadership transition.
- Directed a 50+ person consulting org spanning AppSec, AI, blockchain, and cryptography.
- Managed P&L (\$28M revenue); optimized budget allocations for conferences, offsites, hiring, and R&D while meeting utilization targets.
- Cut client escalations 50% in 3 months, stabilized VIP accounts, and improved project management with a new escalation framework.

Include Security

Senior Security Assessment & Research Consultant

June 2021 - July 2022

- Built our staff augmentation service offering, embedding with client product security teams to increase capacity and accelerate impact.
- Delivered threat models and security reviews for SaaS, fintech, and autonomous vehicle companies.
- Advanced company brand, operations, and research portfolio through tooling, security finding templates, and vulnerability research.

Montana State University

Assistant Professor of Computer Science

August 2019 - August 2021

- Secured \$3.7M+ in DHS and NSF research funding related to cybersecurity and education as Co-PI.
- Designed and taught undergraduate and graduate courses in operating systems and security (e.g., systems, networks, cryptography).
- Led a research group exploring adversarial ML, secure systems, wireless security, and foundational computing education.

Dartmouth

Security Researcher & Instructor

September 2013 - August 2019

- 10+ publication from research in security/privacy for IoT, mobile health, trusted computing, and wireless anomaly detection.
- Built secure, low-power wearable platform; published peer-reviewed research and open-sourced systems.

Intel Labs, Security & Privacy Research

Visiting Security Researcher

2015, 2016

Attachmate

DevOps Engineer

August 2012 - August 2013

EDUCATION

- Ph.D., Computer Science, Dartmouth, Hanover, NH2013 - 2019
- B.S., Mathematics & Computer Science, Western Washington University, Bellingham, WA2008 - 2012

SELECTED PROJECTS

Security Reviews. Disclosed hundreds of security findings spanning numerous web and mobile apps, cloud native tools, AWS and GCP infrastructure and IaC, browsers, AI applications and services, distributed systems and databases, payment systems, and cryptocurrency systems.

Mobile & IoT Security Research. Co-designed and implemented novel secure, low-power wearable platform; published at [SenSys'16](#) and [WMMADD'14](#), and open-sourced our system, [Amulet](#). Dove-tailed into applied security research on popular consumer devices (Bluetooth, Wi-Fi, SGX, wearable and mobile devices) published at [WiSec'21](#), [MobiCom'19](#), [MobiSys'19](#), [HASP'18](#); Intel [patent'19](#).

SKILLS

Leadership: Org strategy, P&L, hiring, team development, cross-functional alignment, sales and marketing enablement.
Technical: *Languages:* Python, Go, TypeScript, JavaScript, C, Java, Bash, Rust. *Frameworks, Libraries, Web, APIs:* FastAPI, Pydantic, Pydantic AI, SQLAlchemy, pytest, celery, asyncio; Node.js, Prisma, React, Angular; MongoDB, PostgreSQL, MySQL, Redis; REST, GraphQL *Cloud & Infrastructure:* Terraform, AWS, GCP, Kubernetes, Docker, GitHub Actions. *Analysis:* static analysis (e.g., semgrep, objdump, binary ninja), dynamic analysis (e.g., gdb, strace, ptrace, perf, valgrind), network analysis (e.g., Burp Suite, Wireshark, tcpdump, IDA, Binary Ninja, Frida).

AWARDS, LEADERSHIP, AND VOLUNTEER EXPERIENCE

- Board Member, Tech Chair, Greenplay Northwest (nonprofit outdoor childhood education program)2024 - Present
- Research Advisor, Montana State University2019 - 2023
- Volunteer Coach, Hanover High School Track & Field and Couch-to-5k, Hanover, NH2016 - 2019
- Teaching Awards, Dartmouth2013, 2014, 2015